URBANYX

Urban Spatial Analysis
Comprehensive Report

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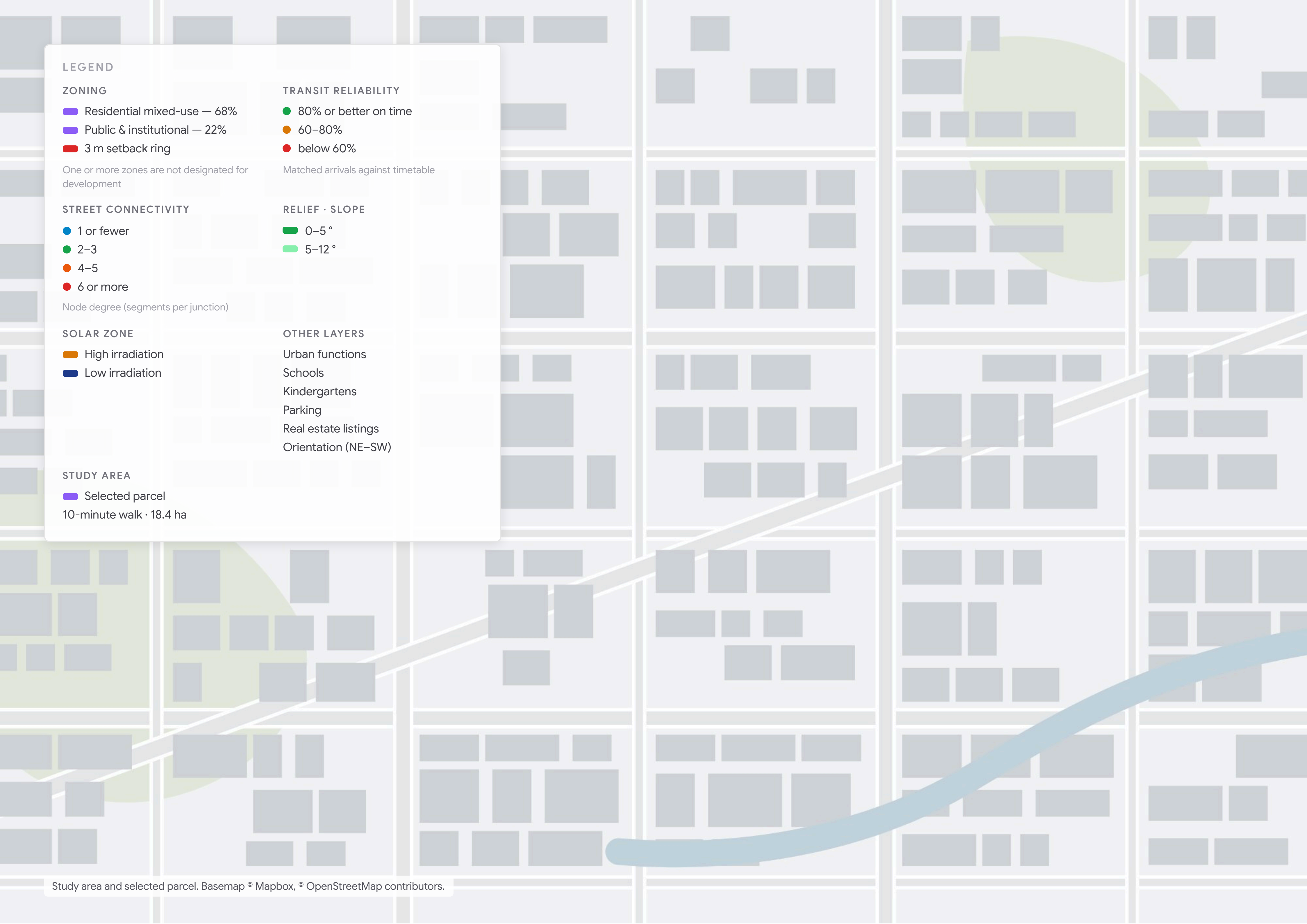
2026

Z·axis

escaping flatland



Study area and selected parcel. Basemap © Mapbox, © OpenStreetMap contributors.

Summary

At a glance, followed by the reasoning behind the figures.

74/100 LIVABILITY Grade B	**₾3,840** FLAT PRICE median / m²	**23%** TREE CANOPY of study area
34.6°C SURFACE TEMP mean	**2,480 m²** PARCEL AREA 01.14.09.023	

This parcel scores 74 out of 100 (grade B) on the Urban Livability Index.

74
Urban Livability Index

B
good

Component	Score	Weight
Education	82	weight 1.0
Transit	71	weight 1.0
Parking	58	weight 0.75
Diversity	76	weight 1.0

How to read this: the index blends walkable access to amenities, public transport, land-use diversity and on-street parking into one 0–100 score (A best, F worst). Components are weighted as shown and averaged over those with data. Definitions are in Methodology.

Findings

Only the analyses active in this export are reported.

Ownership

Parcel code	01.14.09.023
Area	2,480 m²
Parcel type	Non-agricultural
Ownership type	Private
Address	Chavchavadze Avenue 37, Vake, Tbilisi
Registered	14 March 2019
Owner(s)	Kartuli Development LLC

Kartuli Development LLC · ID 405123456 · legal entity

Zoning & development limits

This parcel spans multiple zoning categories. Parameters are calculated per zone, proportional to that zone's share of the parcel area, then summed.

ZONE	SHARE	K1	K2	K3	FOOTPRINT	FLOOR AREA	GREENING	HEIGHT
Residential mixed-use	68%	0.50	2.50	0.20	843	4,215	337	5 fl
Public & institutional	22%	0.40	1.60	0.25	218	874	137	4 fl
Total (m^2)					**1,061**	**5,089**	**474**	**4 fl**

3 m setback ring: 312 m^2 (13% of parcel) — excluded from the buildable footprint above. One or more zones carry K1 = 0; that portion is not designated for development.

Parcel compliance — Article 16

Minimum parcel area	required 500 m^2	actual 2,480 m^2	✓ Meets
Minimum width	required 18 m	actual 41.3 m	✓ Meets
Minimum depth	required 25 m	actual 22.8 m	✗ Below
Maximum building height	limit 5 floors	—	

Dimensions are taken from the minimum-area oriented bounding rectangle of the parcel geometry; requirements per the Tbilisi Land-Use Master Plan, Article 16.

Construction permits

Latest application	02-24-1187 (of 2 on this parcel)
Address	Chavchavadze Ave 37
Nomenclature	Multi-apartment residential building, 5 floors
Registered	11 Feb 2024
Issued	3 Jun 2024
Result	Approved with conditions

⚠ **Flag —** this permit appears to conflict with the Tbilisi 2019 Urban Masterplan: the permitted use is not among those allowed in this zone. Verify against the current masterplan before relying on it.

Street network & morphology

Connectivity	412 segments · 38.2 km · mean node degree 3.00, max 3
Orientation	1204 street ways, dominant axis NE–SW
Urban functions	Retail 31% · Food 22%

Relief · slope · aspect

Active layer	Slope
Statistics	mean 9.4° · max 34.1°

Energy potential

Solar	41% above 4 kWh/m²/day
Wind	3.4 m/s mean · 118 W/m² · 6,200 kWh/yr (5 kW ref.)

Climate & land cover

Tree canopy	23% covered
Land surface temperature	34.6°C mean

Mobility & access

72% on time
Matched arrivals, 30 days

3 areas · ~210 cars
On-street parking capacity

Transit reliability	72% on-time, 26% late (>5 min) across 8,302 matched arrivals at 23 stop-route pairs
Parking split	2 free · 1 paid; 210 car spaces, 4 taxi, 2 loading

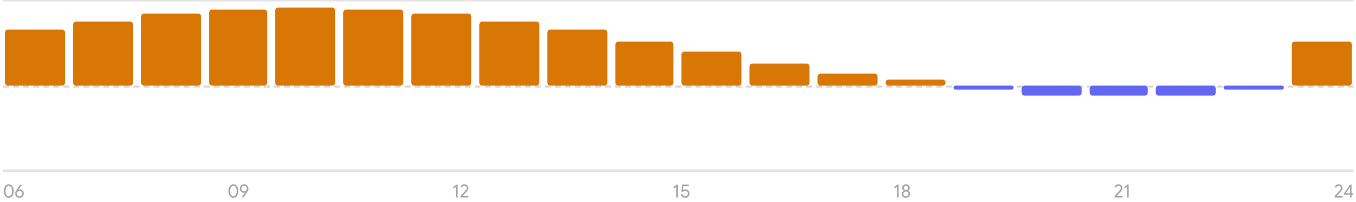
Transit reliability

B Area reliability across every stop in the catchment, weighted by matched arrivals. **72%**

72%	+1.7 min	+9.9 min	+4.1 min
ON TIME	MEDIAN DELAY	90TH PERCENTILE	EXCESS WAIT
of matched arrivals	across stops	across stops	observation-weighted

Window	2026-08-07 → 2026-09-05
Filters	30 days · All days · All day
Archive	since 1 June 2026 · 97 days
Sample	8302 matched arrivals from 13282 observations at 23 stops

Delay by hour of service



Median delay per service hour. Above the line is late, below is early. Bars are scaled to the largest absolute value in the window.

Map thresholds

On time share	≥80% · 60–80% · <60%	Late share	≤10% · 10–25% · >25%
Median delay	≤1.5 min · 1.5–4 min · >4 min	Scheduled headway	≤10 min · 10–20 min · >20 min

Least reliable stops

Marjanishvili · 6, 31	37% late · 59% on time · median +3.3 min
Rose Sq · 7, 32	37% late · 59% on time · median +3.3 min
Tamarashvili · 8, 35	25% late · 73% on time · median +1.8 min
Paliashvili · 9, 31	25% late · 73% on time · median +1.8 min
Chavchavadze Ave · 4, 31	24% late · 73% on time · median +1.7 min

Most reliable stops

Freedom Sq · 8, 33	8% late · 92% on time · median −0.5 min
Vera Park · 5, 35	23% late · 75% on time · median +1.5 min
Nutsubidze · 4, 34	23% late · 75% on time · median +1.5 min
Sandro Euli · 4, 35	23% late · 75% on time · median +1.6 min
Kipshidze · 12, 34	23% late · 74% on time · median +1.6 min

Nearby amenities — walkable catchment

Schools	3
Kindergartens	1
Transit stops	11 — 3 routes (4, 12, 37) · reliability B (88% on-time)
Parking areas	3 (2 free · 1 paid) — capacity ~210 cars
Land-use diversity	76/100 (Shannon SDI, normalised)

Real estate market — catchment

PROPERTY TYPE	MEDIAN PRICE / M²	LISTINGS
Flat	₾3,840	284
House	₾3,120	41

392 listings in the catchment, collected over the preceding 30 days. Asking prices, not recorded transactions.

Transit reliability by segment

The archive is re-queried for every combination of period, day type and time band, over the same 23 stops in the catchment. Shaded rows are the all-day, all-week baseline the summary above reports.

Reliability by service segment — 30 days

2026-08-07 → 2026-09-05

DAY TYPE	TIME BAND		ON TIME	LATE	MEDIAN	P90	EXCESS WAIT	HEADWAY	MATCHED
All days	All day	B	72%	26%	+1.7 min	+9.9 min	+4.1 min	17.4 min	8,302
	AM peak	D	59%	39%	+2.1 min	+11.2 min	—	—	1,413
	Midday	B	76%	22%	+1.6 min	+9.5 min	—	—	2,324
	PM peak	D	54%	44%	+2.3 min	+11.7 min	—	—	1,577
	Evening	B	70%	28%	+1.8 min	+10.1 min	—	—	1,163
Weekdays	All day	C	67%	31%	+1.9 min	+10.4 min	+4.1 min	17.4 min	5,977
	AM peak	D	54%	44%	+2.3 min	+11.7 min	—	—	1,018
	Midday	B	71%	27%	+1.7 min	+10.0 min	—	—	1,673
	PM peak	E	49%	49%	+2.5 min	+12.2 min	—	—	1,136
	Evening	C	65%	33%	+1.9 min	+10.6 min	—	—	836
Sat	All day	A	80%	18%	+1.4 min	+9.1 min	+4.1 min	17.4 min	1,246
	AM peak	C	65%	34%	+1.9 min	+10.4 min	—	—	211
	Midday	A	84%	15%	+1.3 min	+8.7 min	—	—	347
	PM peak	D	59%	32%	+2.0 min	+10.9 min	—	—	234
	Evening	B	79%	17%	+1.5 min	+9.3 min	—	—	174
Sun	All day	A	84%	14%	+1.3 min	+8.7 min	+4.1 min	17.4 min	1,079
	AM peak	B	73%	27%	+1.7 min	+10.0 min	—	—	185
	Midday	A	90%	9%	+1.1 min	+8.3 min	—	—	301
	PM peak	C	65%	33%	+1.9 min	+10.5 min	—	—	206
	Evening	A	83%	17%	+1.4 min	+8.9 min	—	—	152

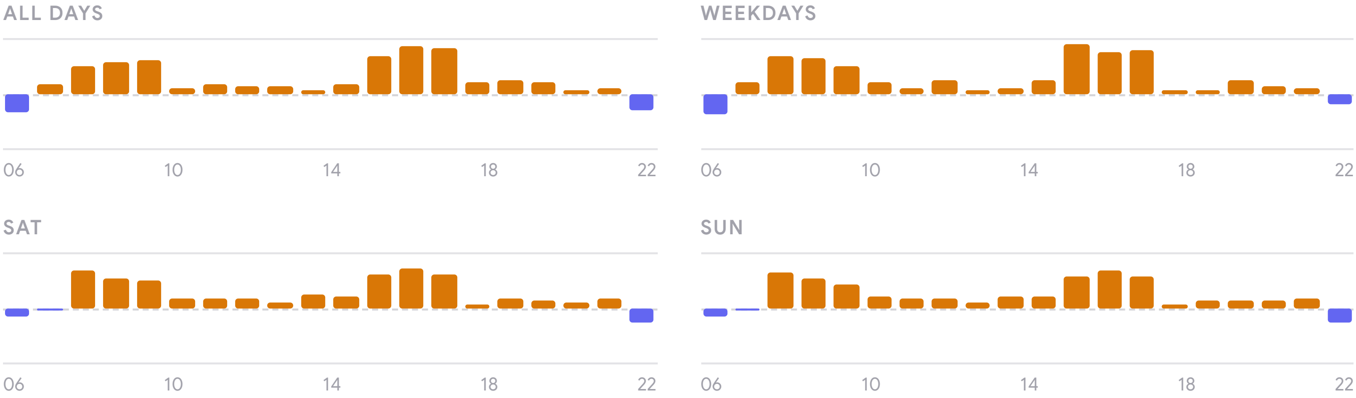
Reliability by service segment — 7 days

2026-08-30 → 2026-09-05

DAY TYPE	TIME BAND		ON TIME	LATE	MEDIAN	P90	EXCESS WAIT	HEADWAY	MATCHED
All days	All day	B	72%	26%	+1.7 min	+9.9 min	+4.1 min	17.4 min	1,939
	AM peak	D	59%	38%	+2.1 min	+11.2 min	—	—	329
	Midday	B	76%	22%	+1.6 min	+9.5 min	—	—	543
	PM peak	D	54%	44%	+2.3 min	+11.7 min	—	—	370
	Evening	B	72%	27%	+1.8 min	+10.1 min	—	—	274
Weekdays	All day	C	67%	31%	+1.9 min	+10.4 min	+4.1 min	17.4 min	1,395
	AM peak	D	57%	43%	+2.3 min	+11.7 min	—	—	235
	Midday	B	70%	26%	+1.7 min	+10.0 min	—	—	391
	PM peak	D	51%	48%	+2.5 min	+12.2 min	—	—	264
	Evening	C	62%	30%	+1.9 min	+10.6 min	—	—	194

DAY TYPE	TIME BAND		ON TIME	LATE	MEDIAN	P90	EXCESS WAIT	HEADWAY	MATCHED
Sat	All day	A	81%	18%	+1.4 min	+9.1 min	+4.1 min	17.4 min	291
	AM peak	D	52%	48%	+1.9 min	+10.4 min	—	—	46
	Midday	A	84%	6%	+1.3 min	+8.7 min	—	—	82
	PM peak	D	50%	50%	+2.0 min	+10.9 min	—	—	48
	Evening	A	96%	4%	+1.5 min	+9.3 min	—	—	46
Sun	All day	A	82%	11%	+1.3 min	+8.7 min	+4.1 min	17.4 min	254
	AM peak	D	52%	28%	+1.7 min	+10.0 min	—	—	46
	Midday	A	97%	3%	+1.1 min	+8.3 min	—	—	71
	PM peak	D	52%	48%	+1.9 min	+10.5 min	—	—	46
	Evening	A	95%	5%	+1.4 min	+8.9 min	—	—	37

- **Delay by hour, by day type**



Median delay per service hour. Above the line is late, below is early. All panels share one scale, so their heights are comparable.

Excess wait and scheduled headway are defined over the whole service day only, so they are left blank in the peak, midday and evening rows. “Matched” is the number of observed arrivals that could be paired with a timetabled trip in that segment; segments with no matched arrivals are omitted.

Methodology

Only the analyses active in this export are defined here — nothing else is carried.

Urban Livability Index

A weighted mean of up to four components, each normalised to 0–1 with a saturating curve and reported on a 0–100 scale: education access (weight 1.0), public transport (1.0, moderated by a route reliability factor), land-use diversity (1.0) and on-street parking (0.75). Weights are renormalised over whichever components have data. Grades: A 80+, B 70–79, C 60–69, D 50–59, E 40–49, F below 40. Counts saturate rather than scale linearly, so a fifth school adds less than a second one.

Walkable catchment

A street-network isochrone around the site (Mapbox Isochrone API) — real walking distance, not a straight-line radius. Large parcels and drawn areas of interest are analysed over their own geometry; running the accessibility isochrone replaces the default catchment with the wider one.

Land-use diversity

Shannon diversity index over amenity-category shares in the catchment, normalised to 0–100. Higher values indicate a more even mix of functions, not more of them.

Transit reliability

The share of arrivals within −60…+300 s of schedule across the catchment's stops, weighted by observation count, over roughly 30 days of archived vehicle positions sampled every 2 minutes. Stops with fewer than 30 matched observations are excluded. Graded A 80%+ down to F below 40%.

Real estate market

Median asking prices for the selected property type across the catchment. Sale medians are per m^2; rent medians are the full monthly price. These are asking prices, not recorded transactions.

Zoning coefficients

K1 is the maximum building footprint ratio, K2 the maximum floor-area ratio and K3 the minimum greening ratio, each defined per functional zone. Where a parcel spans several zones each coefficient is applied to that zone's share of the parcel area and the results summed. Indicative height is K2 ÷ K1, rounded down. Statutory minimums for area, width and depth follow the Tbilisi Land-Use Master Plan, Article 16; width and depth are measured from the parcel's minimum-area oriented bounding rectangle, not its street frontage.

Relief

Slope and aspect are derived on the fly from a digital terrain model over the study area. Slope classes are fixed bands, not quantiles, so they are comparable between reports.

Solar potential

Irradiation modelled from the terrain's slope and aspect against the sun path for this latitude. It describes the ground and roof surface, and accounts for neither shading by adjacent buildings nor local cloud climatology.

Wind

Mean wind speed and power density at the study area from Global Wind Atlas / Open-Meteo reanalysis. Annual yield is a reference figure for a 5 kW turbine and is not a siting assessment.

Tree canopy

Share of the study area classified as tree cover in ESA WorldCover at 10 m resolution. At that resolution individual street trees are not resolved, so dense street planting reads lower than it appears on the ground.

Land surface temperature

Mean land surface temperature from Landsat 8 thermal bands at 30 m, over cloud-free summer passes. This is the temperature of the ground and roof surfaces, which runs warmer than air temperature.

Data availability

These layers rely on external services. When one is unavailable the affected component is omitted and, where applicable, the index renormalises over the components that remain; such omissions are noted in the relevant section.

Sources

Collected in order of appearance; only sources actually used are listed.

Ownership & registration	National Agency of Public Registry (NAPR), Georgia
Zoning	Tbilisi Municipality functional-zone WFS; K-coefficient limits per zone
Statutory requirements	Tbilisi Land-Use Master Plan, Article 16
Construction permits	Tbilisi Architecture Service (docs.tbilisi.gov.ge); ms.gov.ge spatial permit registry
Street network & functions	© OpenStreetMap contributors (Overpass API)
Elevation	Digital terrain model; slope and aspect derived on the fly
Solar	Slope/aspect irradiation model on the DTM
Wind	Global Wind Atlas / Open-Meteo
Tree canopy	ESA WorldCover 10 m (2021)
Land surface temperature	Landsat 8, 30 m resolution
Transit reliability	TTC vehicle positions archived every 2 min; arrivals matched to timetable (on-time −60…+300 s); stops with <30 observations excluded
Parking	Tbilisi Municipality on-street parking dataset
Education facilities	Open municipal datasets
Amenities	Public schools & kindergartens (municipal open data); transit stops & routes (Tbilisi Transport Company); on-street parking (Tbilisi Municipality); land-use POIs © OpenStreetMap via Overpass API
Real estate	Aggregated public listings, 30-day window
Basemap	© Mapbox, © OpenStreetMap contributors
Catchment	10-minute walking isochrone (Mapbox Isochrone API)

Appendix A — Transit stops in the catchment

Every stop inside the isochrone, with its observed arrivals and service levels over the 30 days baseline window.

STOP	ROUTES	OBSERVED	MATCHED	ON TIME	LATE	MEDIAN	P90	EXCESS WAIT	HEADWAY
Marjanishvili	6, 31	928	580	59%	37%	+3.3 min	+14.3 min	+6.0 min	25.3 min
Rose Sq	7, 32	925	578	59%	37%	+3.3 min	+14.3 min	+6.0 min	25.3 min
Tamarashvili	8, 35	589	368	73%	25%	+1.8 min	+10.1 min	+4.0 min	17.9 min
Paliashvili	9, 31	586	366	73%	25%	+1.8 min	+10.1 min	+4.0 min	17.8 min
Chavchavadze Ave	4, 31	576	360	73%	24%	+1.7 min	+10.0 min	+3.9 min	17.6 min
Vake Park	5, 32	573	358	73%	24%	+1.7 min	+9.9 min	+3.9 min	17.5 min
Melikishvili	6, 33	570	356	74%	24%	+1.7 min	+9.9 min	+3.9 min	17.4 min
Rustaveli Sq	7, 34	566	354	74%	24%	+1.7 min	+9.8 min	+3.8 min	17.4 min
Abashidze	10, 32	582	364	73%	24%	+1.8 min	+10.1 min	+3.9 min	17.7 min
Barnovi	11, 33	579	362	73%	24%	+1.7 min	+10.0 min	+3.9 min	17.6 min
Vazha-Pshavela	5, 31	557	348	74%	24%	+1.6 min	+9.7 min	+3.8 min	17.1 min
Mtatsminda	6, 32	560	350	74%	24%	+1.6 min	+9.8 min	+3.8 min	17.2 min
Univerditeti	7, 33	563	352	74%	24%	+1.6 min	+9.8 min	+3.8 min	17.3 min
Delisi	8, 34	566	354	74%	24%	+1.7 min	+9.8 min	+3.8 min	17.4 min
Saburtalo	9, 35	570	356	74%	24%	+1.7 min	+9.9 min	+3.9 min	17.4 min
Medical Univ	10, 31	573	358	73%	24%	+1.7 min	+9.9 min	+3.9 min	17.5 min
State Univ	11, 32	576	360	73%	24%	+1.7 min	+10.0 min	+3.9 min	17.6 min
Bagebi	12, 33	579	362	73%	24%	+1.7 min	+10.0 min	+3.9 min	17.6 min
Kipshidze	12, 34	550	344	74%	23%	+1.6 min	+9.7 min	+3.7 min	17.0 min
Sandro Euli	4, 35	547	342	75%	23%	+1.6 min	+9.6 min	+3.7 min	16.9 min
Nutsubidze	4, 34	531	332	75%	23%	+1.5 min	+9.4 min	+3.6 min	16.6 min
Vera Park	5, 35	534	334	75%	23%	+1.5 min	+9.4 min	+3.6 min	16.6 min
Freedom Sq	8, 33	102	64	92%	8%	−0.5 min	+4.1 min	+1.0 min	7.1 min

Ordered by late share. "Observed" counts every vehicle position archived at the stop; "matched" counts those that could be paired with a timetabled trip and are the basis of every share in this table. Greyed rows fall below the 30-matched-arrival floor and are excluded from the area grade and the rankings. The dot repeats the on-time colour the stop carries on the map.